



Group Number	8
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1. Introduction & Objective

This project covered the design, fabrication, and testing of a biomedical signal conditioning PCB using Altium Designer (Student Edition). The board acquires surface electromyography (EMG) signals, processes them using a bandpass and 50 Hz notch filter, and outputs the conditioned signal to an Agilent data acquisition (DAQ) unit for real-time observation.

2. Filter Design & Breadboard Prototype

2.1 Bandpass Filter

An active bandpass filter was designed to pass the EMG frequency band (center frequency 220 Hz). The filter was first built and tested on a breadboard before PCB implementation to verify the frequency response and component values.

2.2 Notch Filter

A twin-T active notch filter centered at 50 Hz was designed and cascaded with the bandpass filter to suppress power-line interference. The combined filter response was verified on the breadboard prototype before finalising the component values for the PCB.

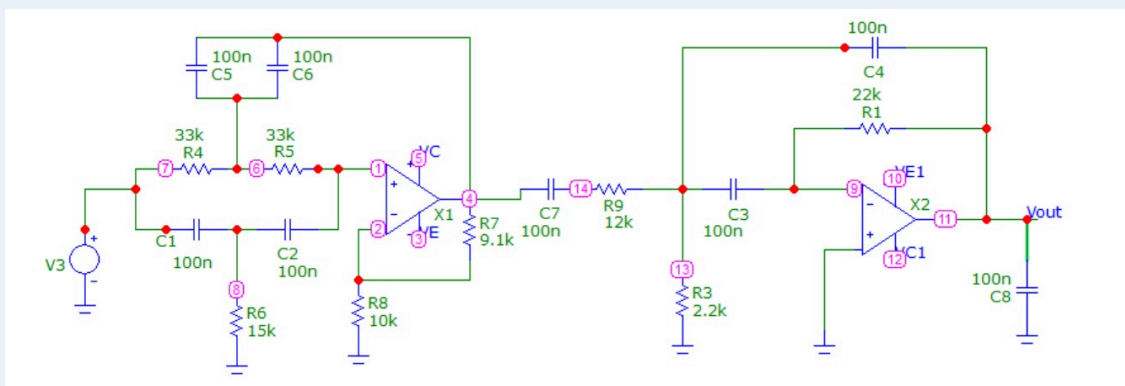


Figure 1. Breadboard prototypes of the bandpass filter and the notch filter.

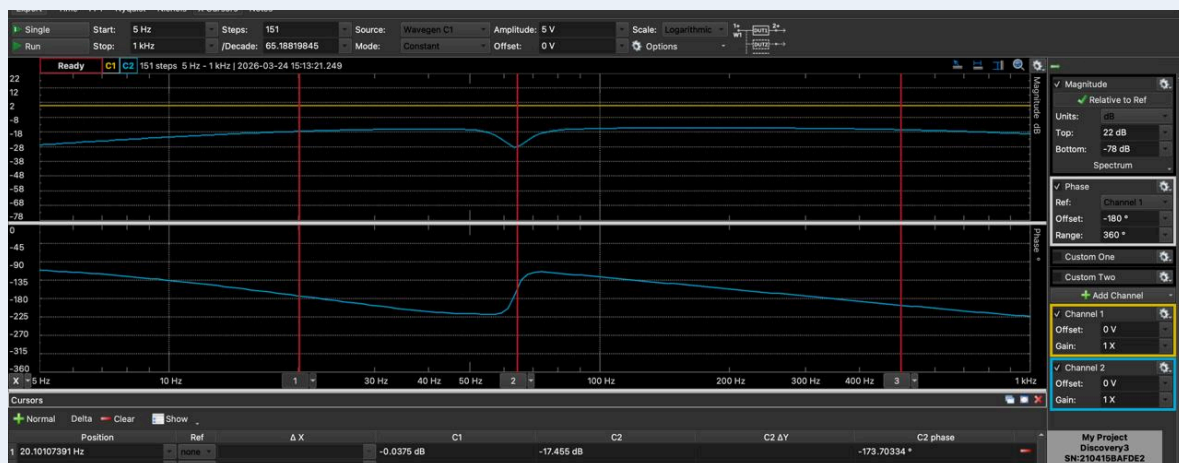


Figure 2. Frequency response of output from bandpass filter and the notch filter.

3. PCB Design in Altium Designer

3.1 Schematic

The provided BioSig schematic (SchDoc) was opened in Altium Designer.

It represents the complete Biomedical Signal Conditioner, integrating the Arduino Uno shield footprint, filter stages, amplifiers, and I/O connectors. The schematic was reviewed and verified before proceeding to PCB layout.

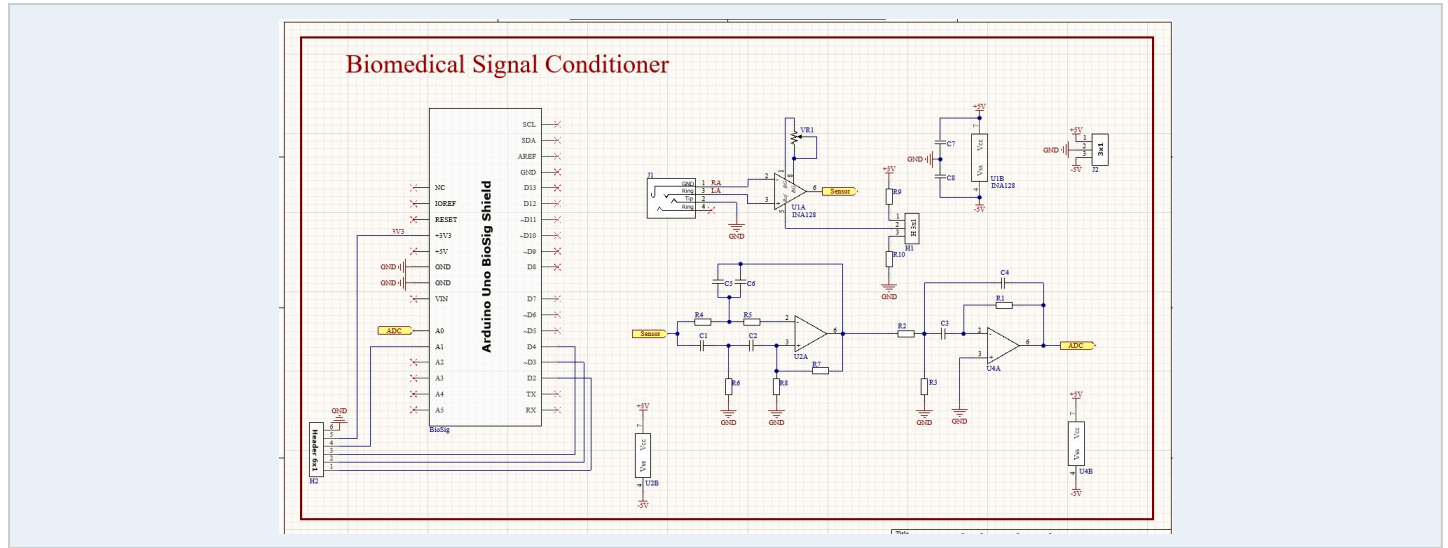


Figure 3. BioSig schematic (SchDoc) in Altium Designer showing the complete circuit.

3.2 PCB Layout & Routing

The BioSig.PcbDoc file was opened along with the schematic. All electrical connections (ratsnest) were manually routed on the two-layer board using the interactive router. Signal traces were placed on the top layer, while ground and power were routed across both layers. Track widths were defined in accordance with the design rules.

After routing, Gerber files and NC drill files were generated via File → Fabrication Outputs and submitted for fabrication.

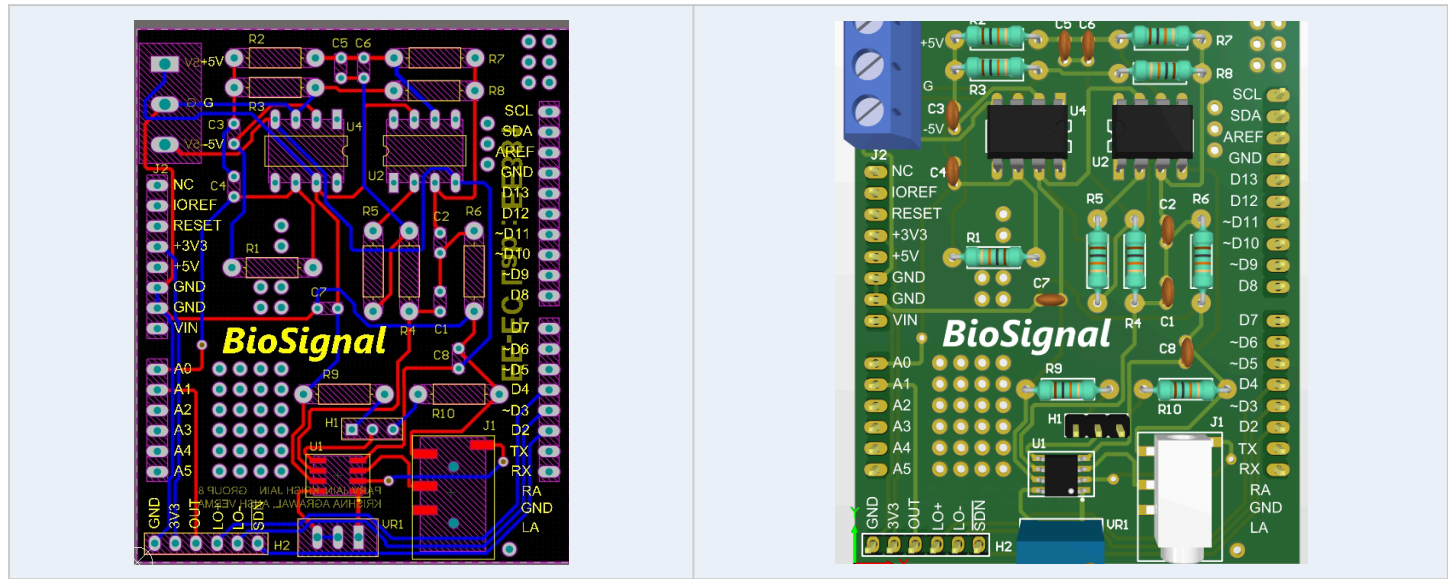


Figure 4. PCB layout after completing all track routes.

4. PCB Fabrication & Assembly

The PCB was fabricated using the generated Gerber files. After receiving the bare board, resistors for the bandpass and notch filter stages were soldered onto the designated pads. Capacitors, ICs, and connector headers were then populated and soldered manually.

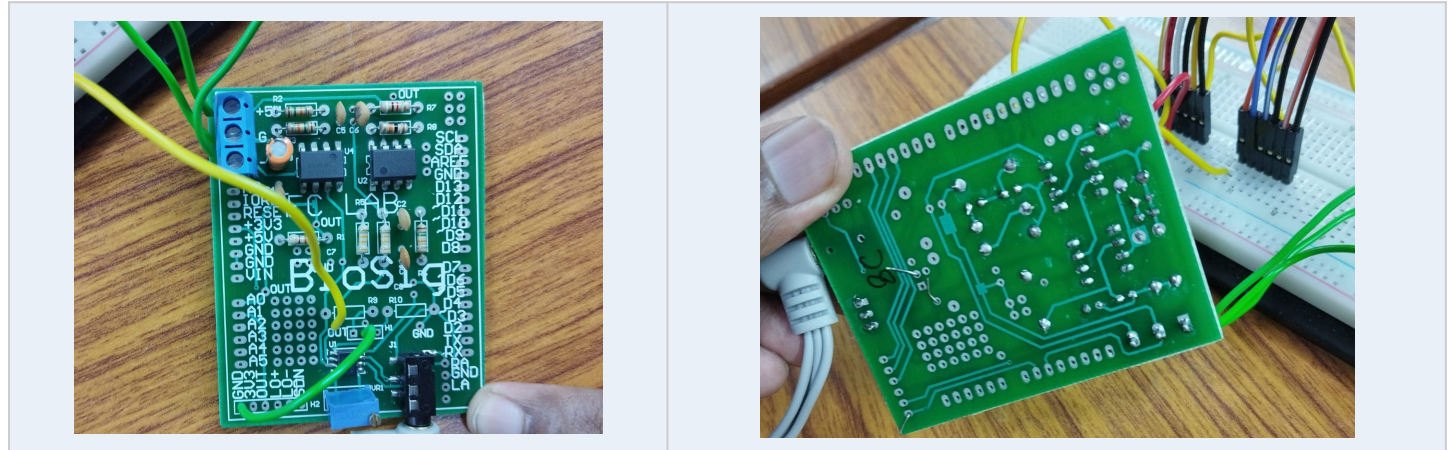


Figure 5. Fully assembled PCB with soldered components.

5. Testing & Results

5.1 Experimental Setup

The EMG probe was placed on the forearm of the test subject. The output of the PCB was connected to the Agilent DAQ input. The subject alternately relaxed and contracted their hand to generate EMG bursts, and the output was observed in real time on the DAQ.

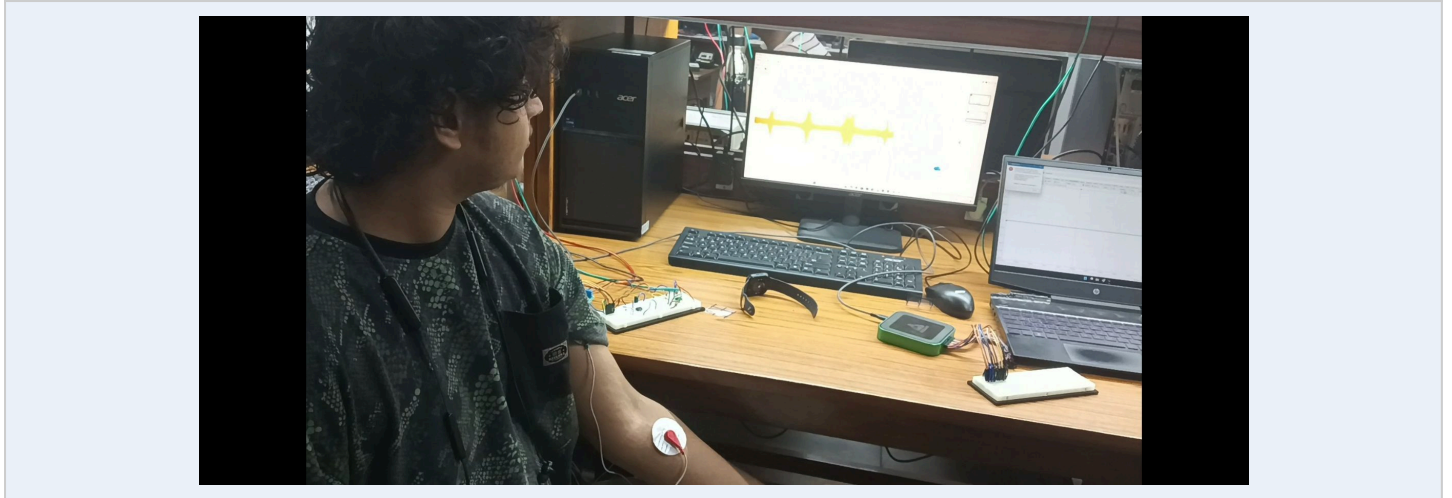


Figure 6. Experimental setup with EMG probe on Krish Jain's hand and output fed into the Agilent DAQ.

5.2 DAQ Output Waveforms

The DAQ waveforms are shown below. In the relaxed state, the output is close to the baseline with minimal noise, confirming that the notch filter effectively suppressed 50 Hz interference. During muscle contraction, clear high-amplitude EMG bursts are visible within the passband, confirming correct operation of the bandpass filter.

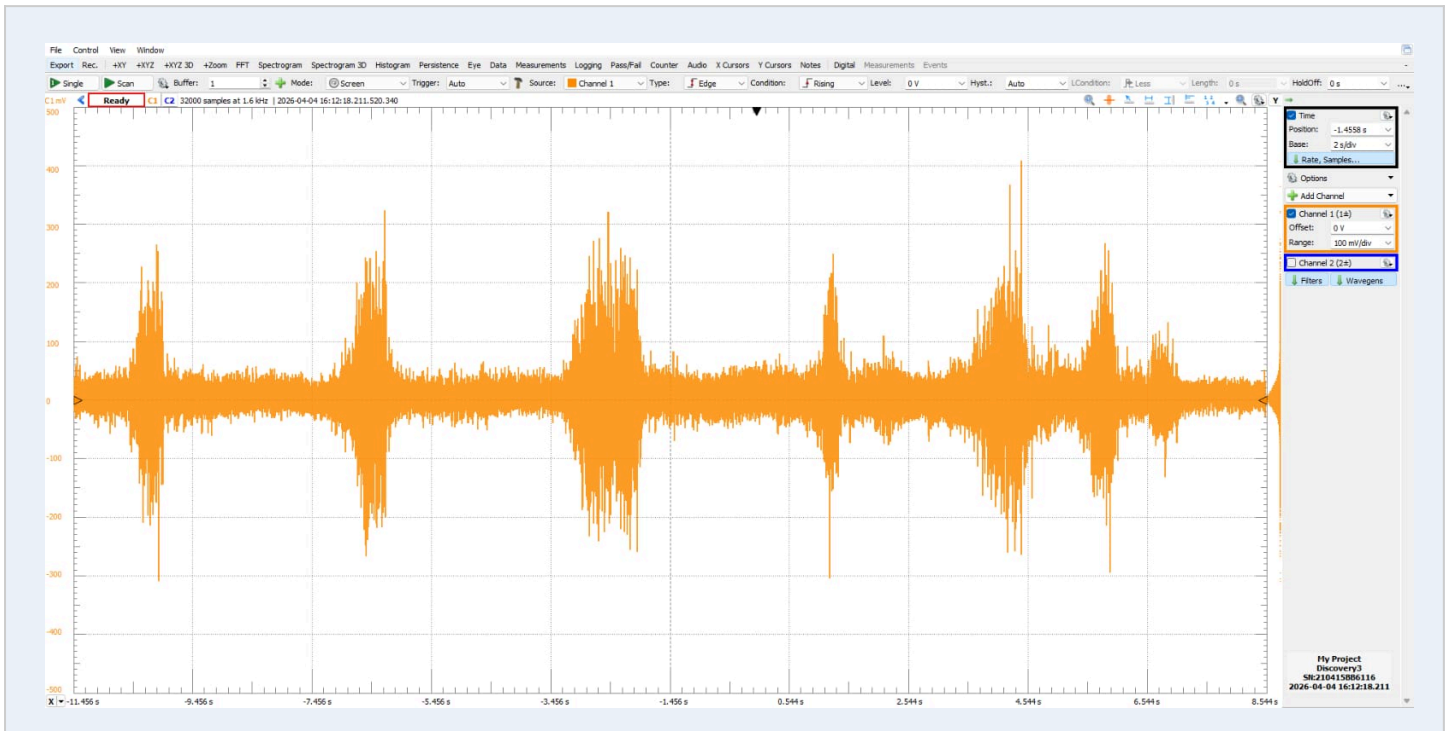


Figure 7. Agilent DAQ output: relaxed and contraction states.

6. Conclusion

The BioSignal Conditioner PCB was designed in Altium Designer, then fabricated, assembled, and tested. The bandpass filter isolated the EMG frequency range, and the notch filter reduced 50 Hz mains interference; both were first verified on a breadboard and then confirmed on the final PCB. EMG signal variations corresponding to voluntary hand contractions were observed on the Agilent DAQ. The project provided hands-on experience with PCB design (schematic capture, routing, Gerber generation), filter implementation, soldering, and basic biomedical signal acquisition.